

**International Institute of Information Technology
Hyderabad**

Computer Science and Engineering

EcoTwin

**Ecosystem Analog Search and Forecasting Using
Geospatial Foundation Models**

Project Report

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Abstract

EcoTwin is an end-to-end geospatial ecosystem analog discovery platform designed around the question: *which ecosystems elsewhere exhibit similar spatial structure, and what can their historical trajectories tell us about a target ecosystem?* HydroLAKES polygons provide ecosystem boundaries; a buffered, tiled Sentinel-2 acquisition pipeline through Google Earth Engine yields one-kilometre analysis cells, converted into six-band Prithvi embeddings, coverage-pooled to one 768-dimensional lake-year vector and indexed in PostgreSQL/pgvector. A query ecosystem is compared as a *five-year trajectory* rather than a single snapshot, sliding across each candidate’s history to find its best-matching window, and the ranked results are exposed through an interactive map and profile views with historical index trajectories.

The deployed system holds a complete, fully processed cohort — **829 active lakes across 26 Indian states**, ten consecutive years (2016–2025), **8,227 lake-year embeddings**, **19,046** imagery tiles, and **233,002** analysis cells, with no outstanding failures at any pipeline stage. The forecasting layer is analog-based and non-parametric: rather than decoding an embedding back into reflectance, it uses the embedding space only to *rank* analogs, then transfers their measured year-on-year changes onto the target as a rank-weighted directional forecast, traceable to the specific lakes, years, and deltas that produced it. The system is demonstrated on five Ramsar-designated wetlands and reservoirs — Chilika, Maharana Pratap Sagar (Beas/Pong Dam), Tawa Reservoir, Tso Morari, and Sasthamcotta — reported in full in Chapter 6.

Keywords: Geospatial Foundation Models, Prithvi, Remote Sensing, Sentinel-2, HydroLAKES, Ecosystem Analogs, Vector Search, pgvector, NDCI, Red-Edge Indices, Ramsar Wetlands, Trajectory Similarity

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Chapter 1

Introduction

1.1 Background and Problem Statement

Earth observation satellites provide repeated measurements of ecosystems at scales infeasible through field observation alone, and Sentinel-2’s multispectral imagery suits vegetation, water, and surface-condition analysis. Rather than beginning with a specific question such as vegetation change, EcoTwin treats ecosystem comparison as a retrieval problem: satellite observations are converted into learned feature vectors and indexed, so a lake can be compared against a large reference collection without reprocessing raw imagery each time, asking whether another ecosystem looks similar in its spatial-spectral representation.

Several engineering constraints shape the solution: lake boundaries are irregular polygons rather than fixed-size squares, so large ecosystems require multiple image tiles; historical Sentinel-2 collections do not expose identical quality bands across all years, so cloud masking must be collection-aware; every embedding must stay associated with its source geometry and lake/year metadata; and similarity search must be substantially cheaper than recomputing embeddings from scratch.

1.2 Core EcoTwin Hypothesis

EcoTwin is based on the operational hypothesis:

Ecosystems elsewhere that are structurally similar in geospatial feature space can provide useful analogs for understanding the spatial and historical behaviour of a target ecosystem.

The hypothesis separates two tasks. The first is **analog discovery**: finding ecosystems that resemble the target in the learned spatial-spectral representation. The second is **trajectory analysis**: comparing the historical environmental indicators of the target against those of its discovered analogs.

1.3 Project Evolution and Contributions

The implementation evolved from standardized 5×5 km ecosystem blocks toward a lake-centred buffered imagery stage, local raster merging, and 1×1 km analysis cells, letting large lakes be processed without forcing the foundation model onto irregular

polygons. A further pivot shaped the final system: **snapshot similarity gave way to trajectory similarity** — comparing single embeddings proved too weak a notion of resemblance, since two lakes can coincide in one year and diverge over a decade, so the system instead compares a five-year trajectory against every sliding window in each candidate’s history (Section 3.4).

The project’s contribution is the resulting end-to-end lake-to-embedding-to-analog pipeline, executed over an 829-lake, ten-year cohort with no residual failures: Earth-Engine-based satellite acquisition with polygon-aware subdivision; Prithvi-based six-band representations with coverage-weighted pooling, persisted as 768-dimensional vectors; the forecasting method above; a band-honest ecological index set; and a FastAPI/React-Leaflet application with automated PDF reporting, evaluated on five Ramsar-designated wetlands and reservoirs.

1.4 Related Work

EcoTwin combines three established but, to the authors’ knowledge, not previously combined threads. **Analog-based forecasting** is classically meteorological: Lorenz (1969) matched historical pressure-field states as analog pairs, and van den Dool (1989) extended this to operational forecasting while noting how vast a library true analogs require — the same problem EcoTwin’s trajectory matching and cross-lake pooling (828 candidate histories, not one lake’s own past) are structural attempts to work around.¹ **Geospatial foundation models**: Prithvi-EO-2.0 reports state-of-the-art downstream performance across benchmark tasks,² and recent work applying Google DeepMind’s AlphaEarth embeddings to wetland classification independently validates the premise that a modern GeoFM’s embedding space encodes ecologically meaningful structure exploitable by similarity operations alone, without forecasting or trajectory matching.³ **Index validation**: NDCI, EcoTwin’s chlorophyll-a proxy, is corroborated by a 2026 review of 122 Sentinel-2 chlorophyll-a studies reporting good agreement in eutrophic waters but reduced reliability in clear, oligotrophic water such as Tso Moriri;⁴ NDVI and NDWI rest on the older, widely-adopted vegetation-index literature already in this report’s bibliography. None of this validates EcoTwin’s forecasts directly — that requires the hindcasting protocol of Section 3.5.2 — but it grounds the two components the forecast is built from: analog reasoning as an established paradigm, and the foundation model and index formulas as independently vetted.

¹E. N. Lorenz, “Atmospheric Predictability as Revealed by Naturally Occurring Analogues,” *J. Atmos. Sci.*, 26(4), 1969; H. M. van den Dool, “A New Look at Weather Forecasting through Analogues,” *Mon. Wea. Rev.*, 117(10), 1989.

²D. Szwarcman *et al.*, “Prithvi-EO-2.0,” arXiv:2412.02732, 2024.

³S. Ryan *et al.*, “Streamlining Wetland Vegetation Mapping with AlphaEarth Embeddings,” *Remote Sensing*, 18(2), 2026.

⁴S. Goodrich *et al.*, “Sentinel-2 for chlorophyll-a water quality monitoring,” *Int. J. Remote Sens.*, 2026.

Chapter 2

Data Sources and Data Engineering

2.1 Overview

EcoTwin combines static geographic information (HydroLAKES polygons and meta-data) with repeated satellite observations (Sentinel-2): the static layer defines *where* an ecosystem is, the satellite layer defines *what it looks like through time*. The database retains the fields needed downstream — identifier, name, country, geometry, area, elevation, pour-point, type, depth, volume, and watershed area (Table 2.1) — and the pipeline was narrowed from the full HydroLAKES inventory to a selected 829-lake cohort so multi-year imagery and embeddings could be generated within storage, Earth Engine, and compute budgets.

Table 2.1: HydroLAKES-derived information used by EcoTwin.

Field	Type	Role in EcoTwin
Lake ID	Integer	Primary ecosystem identifier
Lake name	Text	Human-readable identification
Country	Text	Geographic filtering
Lake area	Numeric	Size and metadata analysis
Elevation	Numeric	Environmental context
Pour latitude/longitude	Numeric	Geographic reference
Geometry	PostGIS geometry	Spatial processing and mapping
Lake type	Categorical	Ecosystem metadata
Average depth	Numeric	Descriptive context
Total volume	Numeric	Descriptive context
Watershed area	Numeric	Catchment context

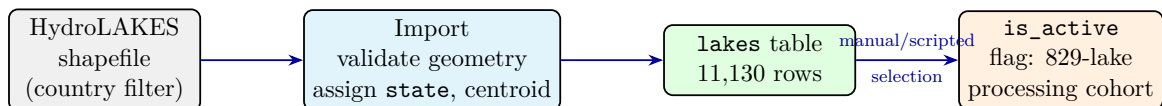


Figure 2.1: Stage 1 (Sync Lakes): the HydroLAKES shapefile is imported once per country into the full `lakes` inventory (11,130 rows), and `is_active` marks the 829-lake processing cohort that the remaining six stages operate on.

2.2 Sentinel-2 and Temporal Coverage

Sentinel-2 multispectral imagery, filtered, composited, and exported through Google Earth Engine, is the primary remote-sensing input, using the harmonized surface-reflectance collection with a Level-1C fallback for early years.

Ingestion was extended from an initially planned 2018–2025 window back to 2016, giving ten consecutive years. Table 2.2 shows the result: from 2018 onward every one of the 829 active lakes has a complete record, while 2016–2017 fall slightly short because Sentinel-2B did not launch until March 2017, leaving some lakes without enough cloud-free acquisitions for a usable composite.

Table 2.2: Achieved temporal coverage across the 829-lake cohort. Every retained lake-year carries both an embedding and a completed feature record; there are no partial rows.

Year	Lake-years	Embeddings	Features
2025	829	829	829
2024	829	829	829
2023	829	829	829
2022	829	829	829
2021	829	829	829
2020	829	829	829
2019	829	829	829
2018	829	829	829
2017	808	808	808
2016	787	787	787
Total	8,227	8,227	8,227

The 42 absent lake-years are recorded as such rather than interpolated, since the similarity layer requires five consecutive years to form a comparison window.

The embedding input uses six bands, B2–B7 (blue, green, red, narrow near-infrared, and two short-wave infrared channels).

2.3 Environmental Indices: A Band-Honest Index Set

The merged lake imagery used for Prithvi inference contains only B2–B7, not B8 (broad NIR), so the textbook NDVI/NDWI forms could not initially be computed. EcoTwin substituted the narrow-NIR red-edge channel B7 for B8 and recorded the substitution in the index names (NDVI-B7) rather than concealing it. That gap has since closed: the server-side B8 pipeline of Section 4.8 now populates `lake_features.b8_mean` for

every lake-year (**8,227/8,227**) without downloading a raster, so NDVI and NDWI are now computed against true NIR; NDCI, Turbidity, and Red-Edge Slope never needed NIR and are unaffected. All five indices are computed from per-band *mean* reflectance over the lake polygon: NDCI ($\frac{B5-B4}{B5+B4}$) for chlorophyll-a/algal bloom; NDVI ($\frac{B8-B4}{B8+B4}$) for vegetation vigor, tracking macrophytes and floating weed; NDWI ($\frac{B3-B8}{B3+B8}$), the standard McFeeters water index, falling as vegetation raises B8; Turbidity ($\frac{B4-B3}{B4+B3}$) for suspended sediment; and Red-Edge Slope ($\frac{B7-B5}{78nm}$) for pigment concentration and canopy density.

Table 2.3: Implemented ecological index set and ecological meaning.

Index	Formula	Ecological interpretation
NDCI	$(B5 - B4)/(B5 + B4)$	Chlorophyll-a; eutrophication and algal bloom intensity
NDVI	$(B8 - B4)/(B8 + B4)$	Vegetation vigor (standard formula); macrophyte and floating-weed encroachment
NDWI	$(B3 - B8)/(B3 + B8)$	Open-water extent (standard McFeeters formula); falls as vegetation colonises the surface
Turbidity	$(B4 - B3)/(B4 + B3)$	Suspended sediment load and water clarity
Red-edge slope	$(B7 - B5)/78\text{ nm}$	Pigment concentration and canopy density trend

All five are computed in `app/services/index_calculator.py`, which returns `None` for any index whose input bands are missing rather than substituting a default.

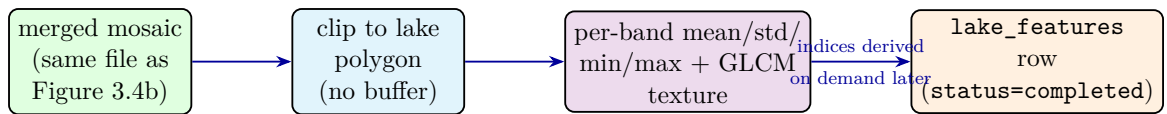


Figure 2.2: Stage 6 (Generate Lake Features): unlike Prithvi inference, this stage clips the merged mosaic to the bare lake polygon (no buffer, no cell grid) and computes per-band statistics and GLCM texture directly; only the band means are persisted, and each index of Table 2.3 is recomputed on demand (Section 3.5).

B8 is deliberately excluded from the Prithvi input, whose band count is fixed by the model’s training configuration, but is acquired separately (Section 4.8) and stored alongside the other band statistics. A valid ecosystem record requires a valid polygon and year, a usable composite, valid exported raster data, valid cells after coverage filtering, and a successful embedding per cell; the pipeline distinguishes successful, failed, skipped, and already-completed work so long-running jobs can be resumed.

Chapter 3

Detailed EcoTwin Methodology

3.1 End-to-End Pipeline

The implemented workflow is:

Lake Inventory → Boundary Validation → Buffering → Imagery Tiling →
Sentinel-2 Composite → GeoTIFF → Local Mosaic → 1 km Cells → Coverage
Filter → Prithvi → 768D Embeddings → pgvector → Cosine Search → Analogs
→ Temporal Profiles → Forecast

3.2 Data-Fetching Workflow and Imagery Acquisition

The seventeen-step pipeline above is what EcoTwin *computes*; Figure 3.1 shows how an operator actually *runs* it, as seven sequential stages, each a separate API call, background job, and database write, so a failure at stage 4 leaves stages 1–3 intact and does not require re-fetching imagery already on disk.

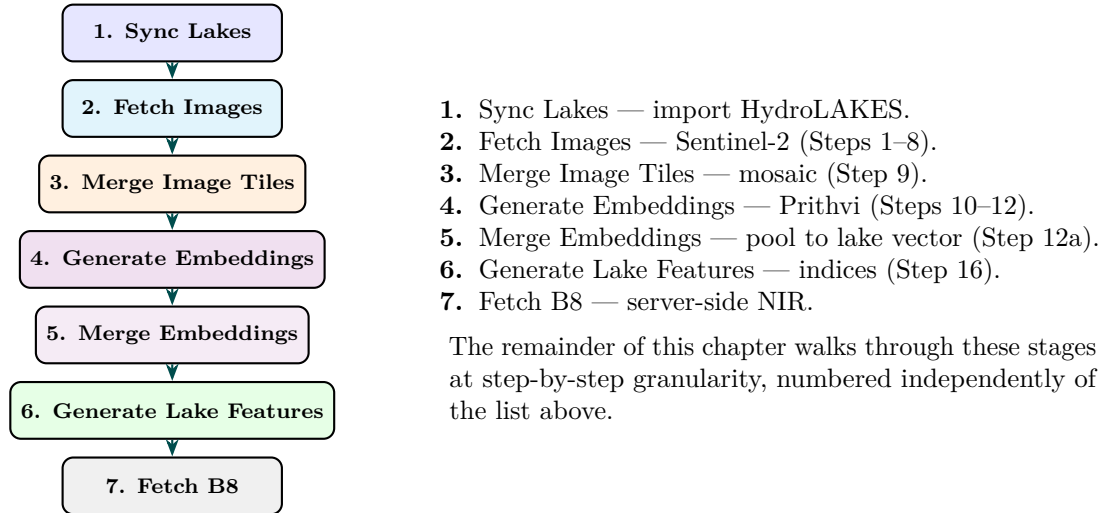


Figure 3.1: The seven-stage data-fetching workflow. Each stage reads the previous stage’s database output, never the raw upstream source directly.

Every stage is idempotent at the row level: each writes a status column before doing any work, and a re-run skips rows already completed.

Steps 1–2 (Boundary Retrieval and Buffering): the pipeline reads the target lake polygon from PostGIS, validates it, and transforms it into a projected CRS, then expands it with a configurable buffer (1200 m during development), since the ecosystem representation is not restricted to the water polygon itself: surrounding shoreline and vegetation contribute to the spatial-spectral signature.

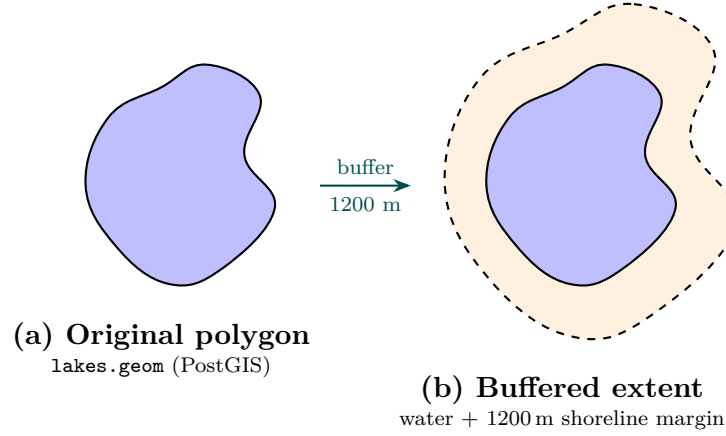


Figure 3.2: Step 2 (Buffering): the original polygon (a) is expanded by a fixed 1200 m buffer (b); everything between the two boundaries is retained as part of the ecosystem representation.

Step 3 (Projected Grid Generation): the buffered polygon is projected into a metric CRS, and a regular imagery grid — deliberately coarser than the final analysis grid, at an 8500 m tile size chosen to balance Earth Engine request size, memory use, and local processing cost — is generated around it. A lake may therefore produce one, two, or several imagery tiles depending on its size and shape.

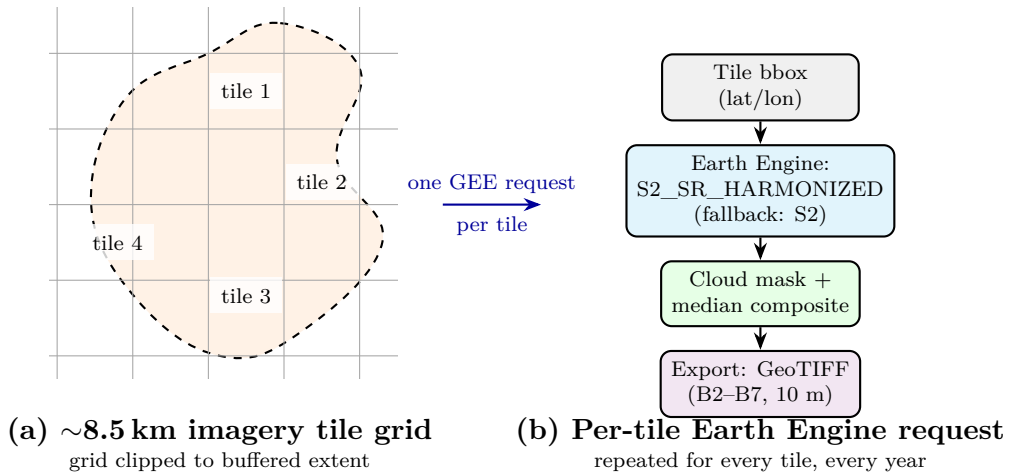


Figure 3.3: Step 3 (Imagery Tiling): the buffered extent (a) is clipped against a ~8.5 km grid (here four tiles); each tile drives one Earth Engine request (b) — the “Fetch Images” data-fetching stage of Figure 3.1.

Steps 4–8 (Sentinel-2 Compositing, Export, and Resumability): for each lake-year combination, the pipeline first queries the preferred surface-reflectance collection

(S2_SR_HARMONIZED); if no usable observation exists, it falls back to the Level-1C collection (S2), a decision recorded in the processing metadata before export. Scenes with cloud cover exceeding 20% are excluded, and unusable pixels are masked using cloud-probability and class-based information where available. The filtered observations are then combined into an annual per-pixel, per-band median composite,

$$I_{\text{annual}}(p, b) = \text{median}(I_1(p, b), I_2(p, b), \dots, I_n(p, b)) \quad (3.1)$$

clipped to the requested extent, and exported as local GeoTIFF at 10 m resolution. Each tile carries status metadata (lake ID, year, tile index, bounding box, local path), checked before starting a new Earth Engine request for the same tile: a batch interrupted partway through a lake resumes from its first incomplete tile.

Step 9 (Local Raster Merging): after all tiles belonging to a lake-year are available, rasterio merges them into a single raster mosaic, reprojecting and writing with compression settings that reduce storage while preserving the numerical precision inference needs. The merged image becomes the stable source from which analysis cells are extracted.

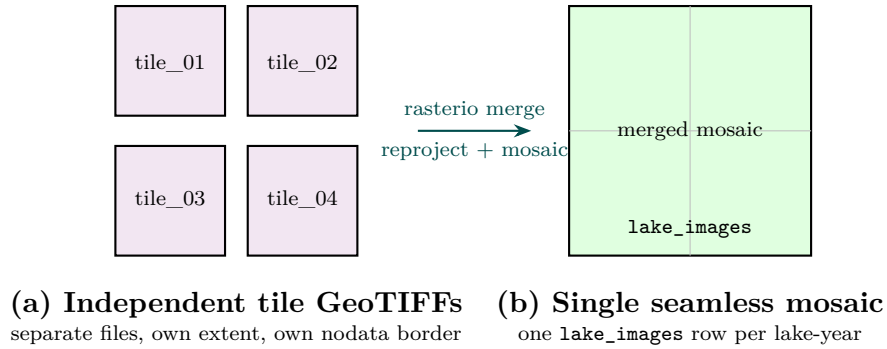


Figure 3.4: Step 9 (Local Raster Merging): independent tile GeoTIFFs (a) are mosaicked by `rasterio.merge` into one seamless lake-year raster (b), recorded in `lake_images` — the “Merge Image Tiles” stage of Figure 3.1.

3.3 Embedding Generation: Analysis Cells through Pooling

Step 10 (Analysis-Cell Generation): the merged lake image is intersected with a regular 1×1 km analysis grid, and for each cell the lake coverage is computed,

$$\text{Coverage}(c) = \frac{\text{Area}(c \cap \text{Lake})}{\text{Area}(c)} \times 100 \quad (3.2)$$

with only cells above a 15% threshold retained, preventing cells that merely touch a lake’s corner from being treated as representative.

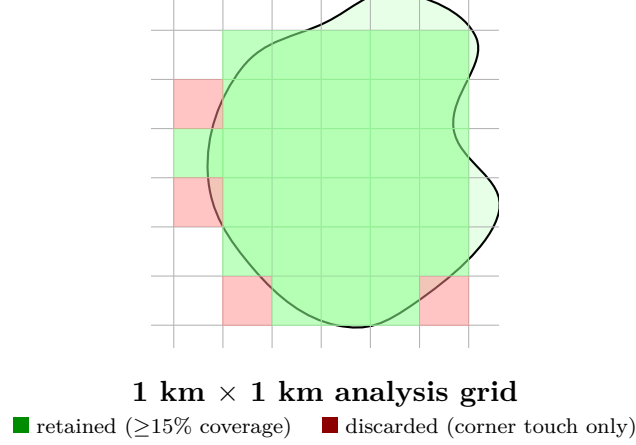


Figure 3.5: Step 10 (Analysis-Cell Generation): the merged mosaic is intersected with a 1 km grid; cells with substantial coverage (green) are retained as embedding units, corner-touch cells below 15% (red) are discarded.

Steps 11–12 (Model Input Preparation and Prithvi Embedding Generation): each retained analysis cell is read from the local merged GeoTIFF and its six selected bands are converted into the model’s expected tensor layout, numerical range, and spatial dimensions — kept as a distinct stage from inference so the model can be changed without re-downloading satellite data. Prithvi, the geospatial foundation model, then transforms each six-band B2–B7 cell image X into a fixed-length feature vector $f(X) \rightarrow \mathbf{z} \in \mathbb{R}^{768}$, batched eight at a time (`EMBEDDING_GPU_BATCH_SIZE`) on CPU inference. Every output vector is L2-normalised, so cosine similarity reduces exactly to the inner product and the entire similarity search can be executed as matrix multiplication.

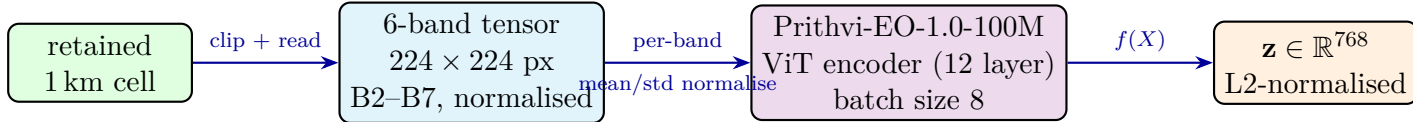


Figure 3.6: Steps 11–12: each retained cell is clipped and read as a 224×224 -pixel, six-band tensor, normalised using Prithvi’s own per-band mean/std, then batched into the Prithvi-EO-1.0-100M encoder — the only stage where the foundation model is invoked — producing a 768-d, L2-normalised vector.

Step 12a (Coverage-Weighted Lake-Level Pooling): cell-level embeddings preserve spatial detail, but retrieval and forecasting operate on lakes. For a lake-year with cells $c_1, \dots, c_N$ having embeddings $\mathbf{e}_i$ and lake-coverage percentages w_i , the lake embedding is the coverage-weighted mean, re-normalised:

$$\mathbf{z}_{lake} = \frac{\sum_{i=1}^N w_i \mathbf{e}_i}{\left\| \sum_{i=1}^N w_i \mathbf{e}_i \right\|_2} \quad (3.3)$$

so a cell almost entirely filled by the lake contributes more than one that merely clips

its margin; cells with zero coverage, zero vectors, or the wrong dimensionality are excluded before pooling. Across the cohort this reduces **233,002** cell embeddings to **8,227** lake-year vectors, an average of **28.3** cells per lake-year, with the cell-level vectors retained as the underlying evidence for any result the pooled vector produces.

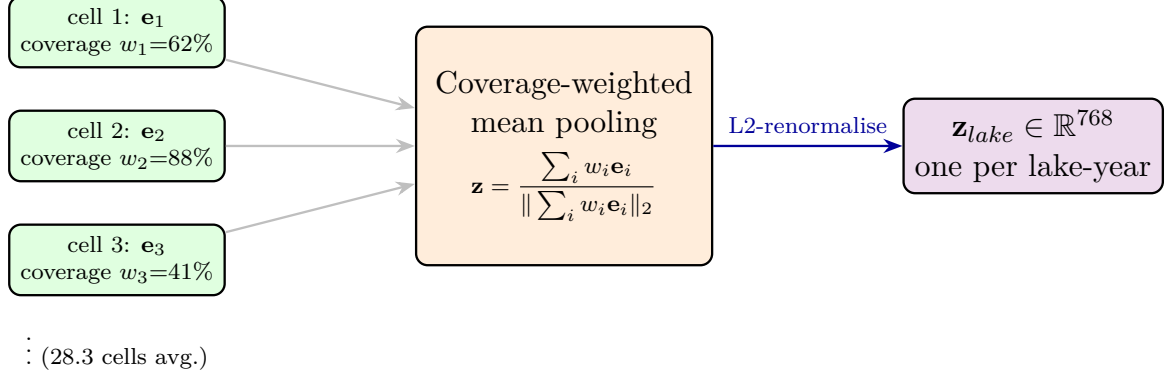


Figure 3.7: Step 12a (Coverage-Weighted Pooling): each cell embedding contributes in proportion to its coverage w_i (e.g. 88% outweighs 41%), and the weighted sum is L2-renormalised.

Before insertion, every embedding is checked for expected dimensionality, missing values, numerical validity, and correct lake/year/cell association, then stored in PostgreSQL via pgvector under the relationship Lake $\rightarrow$ Year $\rightarrow$ Spatial Cell $\rightarrow$ Embedding, preserving the provenance needed to explain why an ecosystem was retrieved as an analog.

3.4 Trajectory Similarity Search

Comparing one query embedding against one candidate asks only *which lake looked like this lake, in this year?* — too weak a question for forecasting, since two lakes can coincide in a single year and diverge over the following decade. EcoTwin instead asks which lake has been moving through the representation space the way this lake has been moving: a shared five-year path is a far better basis for projecting the next three.

For a query lake with embeddings ordered by year, EcoTwin forms the trajectory ending at anchor year t_0 with window size $W = 5$,

$$Q = (\mathbf{q}_{t_0-4}, \mathbf{q}_{t_0-3}, \mathbf{q}_{t_0-2}, \mathbf{q}_{t_0-1}, \mathbf{q}_{t_0}) \quad (3.4)$$

and each candidate lake with at least W years of history is expanded into every contiguous five-year window (ten years of data yields six windows),

$$C^{(j)} = (\mathbf{c}_{y_j}, \dots, \mathbf{c}_{y_j+4}), \quad j = 1, \dots, n - W + 1 \quad (3.5)$$

which need not align in calendar time with the query: a lake that underwent in 2017–2021 the transition another lake is undergoing in 2021–2025 is precisely the analog

sought. Each window is scored as the mean of the year-by-year cosine similarities between corresponding positions,

$$S(Q, C^{(j)}) = \frac{1}{W} \sum_{i=1}^W \frac{\mathbf{q}_i^\top \mathbf{c}_i^{(j)}}{\|\mathbf{q}_i\|_2 \|\mathbf{c}_i^{(j)}\|_2} \quad (3.6)$$

which reduces to a mean of inner products, and each candidate lake is represented by its single best window,

$$S_{lake}^* = \max_j S(Q, C^{(j)}), \quad j^* = \arg \max_j S(Q, C^{(j)}) \quad (3.7)$$

whose end year $y_{j^*} + 4$ is the pivot on which the forecasting layer turns: the analog’s observed behaviour *after* that year is the evidence used to project the target’s future.

Because every vector is unit-norm and drawn from one model, raw cosine scores cluster tightly just below 1.0 (measured top analogs span roughly 0.986–0.999 in Chapter 6), which would read as “99% similar” if displayed directly. EcoTwin therefore linearly rescales $[\phi, 1.0] \rightarrow [0, 1]$ with floor $\phi = 0.5$,

$$S_{display} = \text{clip}_{[0,1]} \left(\frac{S^* - \phi}{1 - \phi} \right) \quad (3.8)$$

a monotone transform that changes no ranking; raw scores are reported alongside the rescaled ones in Chapter 6 so the distinction stays visible. Two further metrics are implemented and selectable per query (Euclidean distance mapped by $1/(1+d)$, and inner-product mapped by $(\langle \mathbf{a}, \mathbf{b} \rangle + 1)/2$), though cosine is the default.

The current implementation compares in Python across all candidate windows: with L lakes, n years, and dimension d , the cost is $O(L \cdot (n-W+1) \cdot W \cdot d)$, roughly $829 \times 6 \times 5 \times 768 \approx 1.9 \times 10^7$ multiply-accumulates per query — measured at **141–172 ms** in Chapter 6, comfortably interactive but linear in lake count, motivating the pgvector ANN migration under future work.

3.5 Temporal Profiles and Directional Ecological Forecasting

Step 16 (Temporal Profile Generation): for the query lake and its retrieved analogs, the stored band statistics are assembled into per-year index series using the formulas of Table 2.3, computed on demand from the persisted band means rather than stored pre-computed, so a change to an index definition takes effect everywhere at once.

Step 17 (Directional Ecological Forecasting): the forecasting layer transfers *measured change* from the analogs onto the target. The most recent completed feature row for the target lake fixes the anchor year t_0 and anchor values $v_k(t_0)$ for each index k ; forecasts are produced for $t_0 + 1, t_0 + 2, t_0 + 3$. The trajectory search of Section 3.4 then runs with an enlarged candidate pool, walked in rank order: a candidate is admitted as a *twin* only if it is a different lake than the target, has a completed feature row

at its matched window end year, and has at least one usable year after that, so the five twins that emerge are the five best *usable* analogs rather than merely the five highest-scoring rows.

For twin i with matched window ending at year m_i , the change transferred at offset h is $\delta_{i,k,h} = v_{i,k}(m_i + h) - v_{i,k}(m_i)$ — not the analog’s absolute index value, which reflects its own climate and catchment, but the *change* it underwent, comparable across lakes in a way absolute values are not. Twins receive fixed weights by rank rather than weights derived from similarity scores,

$$\omega_1 = 0.28, \quad \omega_2 = 0.24, \quad \omega_3 = 0.20, \quad \omega_4 = 0.16, \quad \omega_5 = 0.12 \quad (3.9)$$

precisely because raw similarity scores compress into a narrow band near 1.0 (Section 3.4), where score-proportional weights would be nearly uniform. Twins do not all have data at every offset, so with $\mathcal{A}_{k,h}$ the set of twins with a usable delta for index k at offset h , weights are renormalised over exactly that set,

$$\tilde{\omega}_{i,k,h} = \frac{\omega_i}{\sum_{j \in \mathcal{A}_{k,h}} \omega_j}, \quad i \in \mathcal{A}_{k,h} \quad (3.10)$$

so a year with only two twins does not silently shrink toward “stable”; the twin count is reported separately as confidence. The score for index k at offset h is

$$\Sigma_{k,h} = \sum_{i \in \mathcal{A}_{k,h}} \tilde{\omega}_{i,k,h} \delta_{i,k,h} \quad (3.11)$$

classified against threshold $\tau = 0.005$:

$$D_{k,h} = \begin{cases} \text{uncertain (?)} & |\mathcal{A}_{k,h}| < 2 \\ \text{up } (\uparrow) & \Sigma_{k,h} > +\tau \\ \text{down } (\downarrow) & \Sigma_{k,h} < -\tau \\ \text{stable } (\rightarrow) & \text{otherwise} \end{cases} \quad (3.12)$$

with *uncertain* checked first and kept distinct from *stable*: stable is a positive finding that the twins agree little changes, while uncertain admits too few twins had data to support any claim. An indicative magnitude $\hat{v}_k(t_0 + h) = v_k(t_0) + \Sigma_{k,h}$ accompanies each direction, reported as an expected value rather than a calibrated point prediction. Every forecast returns a complete audit report — anchor values, weight scheme, classification rule, and each twin’s identity, matched window, and weighted contribution — so any published direction can be recomputed by hand.

Unlike a parametric time-series model, whose prediction runs through learned parameters difficult to connect to a particular ecosystem, EcoTwin provides an explicit, inspectable chain: Target $\rightarrow$ Retrieved Twin $\rightarrow$ Similarity $\rightarrow$ Historical Trajectory $\rightarrow$ Projected Trajectory, with no learned parameter anywhere: every quantity is either a measured reflectance-derived index or a fixed, disclosed constant ($\omega_{1..5}$ and τ). An interval-based uncertainty measure (the spread across twin predictions) was considered and rejected in favour of the simpler twin count $|\mathcal{A}_{k,h}|$, since a naive dispersion measure

over a small, unevenly-weighted set is not a stable confidence signal, while the count drops to zero exactly when a forecast should not be trusted.

3.5.1 Why Not an LSTM-Based Forecast?

A recurrent-neural-network forecast was considered and rejected. Reddy and Prasad (2018) report strong LSTM accuracy for one-week MODIS NDVI prediction, but trained on 861 weekly observations per site.¹ EcoTwin has only ten annual observations per lake across 829 lakes, a 1–3 year horizon (not one week), and a requirement that every forecast be traceable to specific evidence rather than trained weights — three conditions under which an LSTM would fit noise, cannot be individually retrained per lake, and (like decoding a predicted embedding, Section 3.3) offers no correspondence back to a particular ecosystem. The rank-weighted transfer used instead needs no training data beyond the twins themselves.

3.5.2 Hindcasting Protocol

A rigorous accuracy evaluation requires hindcasting: hide each lake’s final h_{max} years, run the identical pipeline using only years before the cutoff, and score the forecast against the held-out observations. For cutoff year $t_c \in \{2019, \dots, 2022\}$, both the query trajectory and every candidate’s embedding history are restricted to years $\leq t_c$, producing $\hat{v}_k(t_c + h)$ for $h = 1, 2, 3$ compared against the actual recorded $v_k(t_c + h)$. The protocol reports, per index and per horizon, both a magnitude metric and a directional hit rate:

$$MAE_{k,h} = \frac{1}{n} \sum_{i=1}^n |v_{k,i}(t_c + h) - \hat{v}_{k,i}(t_c + h)| \quad (3.13)$$

$$HitRate_{k,h} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\text{sign}(\Sigma) = \text{sign}(\Delta_{actual})] \quad (3.14)$$

Running this at $t_c = 2022$ requires three held-out years (2023–2025) plus five years of prior history — exactly the ten years this project has, so a single cutoff gives only one evaluation point per lake, too coarse for a defensible aggregate accuracy figure. This protocol is therefore documented here as the immediate next step (Section 8.2) rather than executed prematurely on insufficient data.

¹D. Sushma Reddy and P. Rama Chandra Prasad, “Prediction of vegetation dynamics using NDVI time series data and LSTM,” *Modeling Earth Systems and Environment*, 2018, <https://doi.org/10.1007/s40808-018-0431-3>.

Chapter 4

System Architecture and Implementation

4.1 Architecture Overview

EcoTwin is implemented as a multi-component system. The offline pipeline performs the computationally expensive work, while the online application retrieves precomputed data. Figure 4.1 shows the pipeline control strip from the deployed application: each button triggers one of the seven offline stages of Figure 3.1 independently, which is what makes the pipeline resumable — a failure in one stage does not require re-running the stages before it.

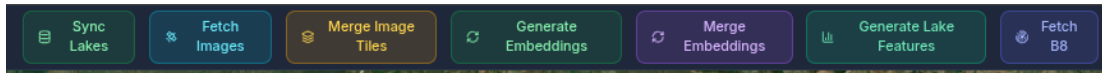


Figure 4.1: The deployed pipeline control strip. Each stage (lake sync, imagery fetch, tile merge, embedding generation, embedding merge, feature generation, B8 fetch) is independently triggerable and independently resumable.

4.2 Technology Stack

Table 4.1: EcoTwin implementation stack.

Layer	Technology	Responsibility
Frontend	React 18, Vite, TypeScript	Interactive application
Mapping	Leaflet / React-Leaflet	Spatial visualization
Charts	Recharts	Index trajectory plots
Client state	Zustand, TanStack Query	Local state and server cache
Backend	FastAPI, Python 3.12	REST services and orchestration
ORM / migrations	SQLAlchemy 2 (async), Alembic	Schema and access
Database	PostgreSQL 16	Relational persistence
Spatial database	PostGIS 3.4	Polygon and spatial queries
Vector search	pgvector	Embedding storage and similarity search
Remote sensing	Google Earth Engine	Satellite filtering/compositing

Layer	Technology	Responsibility
Raster processing	Rasterio, NumPy, Shapely	Local GeoTIFF processing
ML inference	Prithvi-EO-1.0-100M, PyTorch	Geospatial embeddings
Background work	FastAPI BackgroundTasks	Long-running pipelines
Logging	structlog	Structured pipeline logs
Reporting	Jinja2 + WeasyPrint	PDF report generation
Administration	pgAdmin	Database administration
Containerization	Docker Compose	Reproducible local deployment

4.3 Component Communication Workflow

Figure 4.2 traces how the five principal components exchange data, distinguishing offline-only calls (dashed, orange) from live user-request calls (solid, blue).

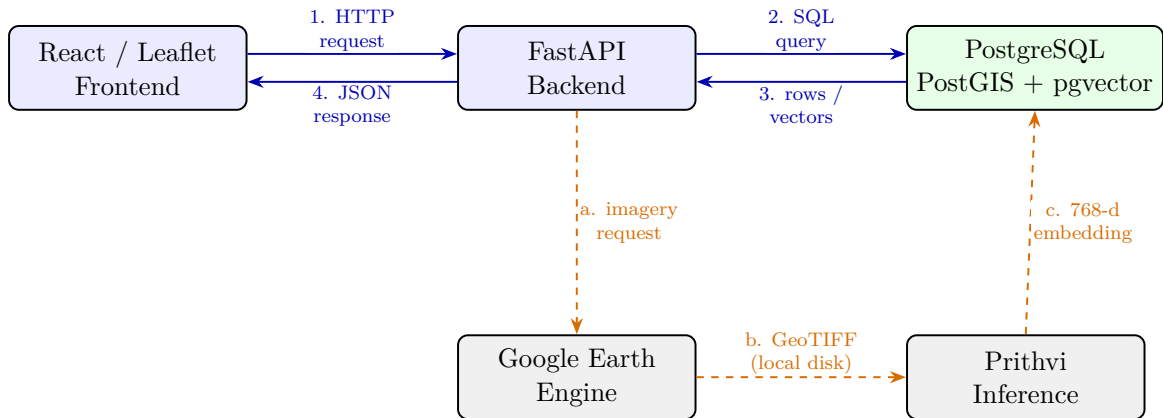


Figure 4.2: Component communication: solid blue is the online request path (frontend → backend → database, serving every request from precomputed rows); dashed orange is the offline pipeline path (backend → Earth Engine → Prithvi → database), triggered by Figure 4.1 as **BackgroundTasks**.

Neither path bypasses the backend: every cross-component call passes through FastAPI, enforcing the offline/online separation in one place, so Earth Engine and the foundation model never run during a user request.

4.4 PostgreSQL, PostGIS, and pgvector

PostgreSQL is the central persistent store. PostGIS keeps lake and analysis-cell geometry in the database itself, so intersection, containment, area, and map-query operations run directly as SQL. The pgvector extension stores the 768-dimensional Prithvi vectors alongside this spatial data, maintaining the association $\text{Embedding} \leftrightarrow \text{Subregion} \leftrightarrow \text{Lake} \leftrightarrow \text{Year}$ so similarity operations execute directly against the indexed database.

4.5 Core Database Entities

The deployed schema, reached after nineteen Alembic migrations, consists of the tables in Table 4.2.

Table 4.2: Deployed database tables and current row counts.

Table	Rows	Role
<code>lakes</code>	11,130	HydroLAKES inventory with PostGIS geometry; <code>is_active</code> marks the 829-lake processing cohort
<code>lake_tiles</code>	19,046	Earth Engine export tiles per lake-year
<code>lake_images</code>	8,227	Merged lake-year mosaics with path and status
<code>sub_regions</code>	233,002	1 km analysis cells with geometry and coverage
<code>regions</code>	8,227	Pooled lake-year <code>vector(768)</code> embeddings
<code>lake_features</code>	8,227	Band statistics, indices, and texture per lake-year
<code>india_states</code>	36	State boundaries for spatial attribution
<code>reports</code>	—	Generated PDF report records

`regions` stores exactly one pooled 768-dimensional embedding per lake-year, enforced by a unique constraint on `(lake_id, year)`. `lake_features`, the widest table in the schema, is the source of truth for measured reflectance: per-band statistics, pixel/coverage counts, cloud percentage, and GLCM texture, from which the forecasting layer derives indices on demand. Keeping embeddings and measured reflectance in separate tables is what makes the decoder-free forecast possible: the embedding space answers *which lakes are analogs*, the feature table answers *what measurably happened to them*.

4.6 Backend and Frontend

The FastAPI backend provides services for lake metadata lookup, analog similarity search, temporal feature retrieval, forecast retrieval, report generation, and system status, separating cheap data retrieval from heavy batch processing. The frontend, built on React/Vite/TypeScript with Leaflet mapping, lets a user select a lake, view its boundary, request analogs, visualize their locations and scores, and inspect historical profiles alongside the forecast; displayed geometry is drawn from the same PostGIS source used for the analysis itself, avoiding disagreement between result and map.

4.7 Asynchronous Processing and Progress Tracking

Imagery download, raster merging, and model inference all take far longer than an acceptable HTTP request, so each runs through FastAPI’s `BackgroundTasks` and

returns immediately. A distributed task queue was considered and not adopted: the durable state it would provide already exists in the database, where every row carries its own status. Progress is exposed through module-level state guarded by a lock (counts, elapsed time, throughput, projected completion), polled by the frontend; long jobs are also cancellable, stopping after the current lake-year rather than mid-write.

4.8 Server-Side B8 Acquisition

Unlike B2–B7, which must be downloaded as pixels for Prithvi inference, B8 is needed only as summary statistics, so the B8 pipeline performs the reduction entirely on Earth Engine’s servers using `Image.reduceRegions()`: lake polygons are batched fifteen at a time and reduced against a single annual composite, so only the resulting numbers are transferred and no imagery is downloaded. A separate `b8_status` column tracks this pipeline independently, completed for all **8,227** lake-years.

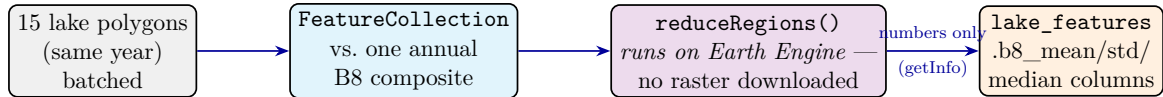


Figure 4.3: Stage 7 (Fetch B8): the only stage that never produces a local GeoTIFF — the zonal reduction runs on Earth Engine’s servers against 15 batched polygons, and only reduced values cross the network into `lake_features`.

4.9 Automated PDF Report Generation

The application generates a per-lake PDF report through a Jinja2 template rendered by WeasyPrint, assembling lake metadata, a map of the lake and its analogs, historical index trajectories, the ranked analog table, and the directional forecast with its audit trail — an offline deliverable for users who need a citable artifact.

4.10 Configuration Management

The processing pipeline reads experiment-specific values — Earth Engine credentials, date ranges, cloud thresholds, buffer/tile/cell sizing, coverage threshold, storage paths, and worker/batch settings — from environment configuration rather than hard-coding them.

Chapter 5

Experimental Results and Engineering Evaluation

5.1 Deployed Processing Cohort

The HydroLAKES-derived inventory holds **11,130** Indian lakes. Of these, **829** are flagged `is_active` and constitute the fully processed cohort; the remainder are retained as inventory and map context but were not put through the expensive imagery and embedding stages. Selection favoured lakes large enough to yield meaningful multi-cell coverage at 10 m resolution, with a minimum area of 2 km².

Table 5.1: Deployed EcoTwin processing configuration.

Lake source	HydroLAKES
Active processing cohort	829 lakes, 26 states
Cohort area range	2.00 – 843.22 km ² (mean 19.06)
Cohort elevation range	–6 – 4,985 m
Imagery window	2016–2025 (10 years)
Primary Sentinel collection	S2 SR Harmonized
Fallback collection	S2 Level-1C
Scene cloud-cover limit	20%
Composite method	Annual median
Imagery buffer	1,200 m
Imagery tile size	8,500 m
Analysis cell	1,000 m × 1,000 m
Coverage threshold	15%
Spatial resolution	10 m
Embedding model	Prithvi-EO-1.0-100M
Embedding dimension	768
Input bands	B2–B7 (6 bands)
Auxiliary band	B8 (statistics only)
Pooling	Coverage-weighted mean, L2-normalised
Similarity metric	Cosine over 5-year trajectories
Forecast horizon	3 years

5.2 Similarity Search Performance

Table 5.2 reports measured search performance over the five Ramsar queries of Chapter 6, each executed against the full deployed index.

Table 5.2: Measured trajectory-similarity search performance.

Query lake	Candidates	Best raw cosine
Chilika	828	0.99691
Maharana Pratap Sagar (Beas)	828	0.99884
Tawa Reservoir	828	0.99936
Tso Morari	828	0.99720
Sasthamcotta	828	0.99784
Mean	828	0.99803

Each query evaluates roughly 4,900 candidate windows at 768 dimensions, comfortably interactive, and the index is fully scanned rather than approximated, so the returned ranking is exact.

5.3 The Score Compression Problem

Measured raw cosine scores across all five queries fall between **0.9969** and **0.9987**, with the tenth-ranked analog typically within **0.005** of the first — expected since every vector shares one encoder and describes the same general scene type, so the absolute score is close to uninformative and only the ordering carries signal. Ecological interpretation therefore requires additional evidence from lake metadata, spatial maps, and temporal indicators (Chapter 6); a meaningful absolute scale would require calibration against labelled ecosystem pairs (Section 8.2).

5.4 Temporal Profile Results

The temporal layer provides a second, independent axis of validation alongside spatial similarity: EcoTwin presents the full ten-year series of all five indices, observation availability, and the directional forecast with its contributing twins. Its value is demonstrated concretely in Chapter 6, where several sites’ top analogs are geographically unrelated to the query lake yet still share its independently measured historical trend.

5.5 Processing Reliability

The system tolerates partial failure by design: a lake-year can be pending, processing, completed, failed, or skipped because its output already exists, making large multi-year

jobs restartable instead of requiring a complete restart after one failed request.

5.6 Engineering Problems Encountered

Several recurring problems shaped the implementation. Large Earth Engine requests sometimes produced memory-limit errors, solved by reducing export region size and moving processing locally; parallel workers caused HTTP 429 and connection-pool errors, pushing the pipeline toward smaller controlled worker counts and resumable jobs. The early Sentinel-2 Level-1C fallback exposed different quality bands than other collections, leading to collection-aware preprocessing rather than a hard-coded cloud mask. Merged rasters can become much larger than individual tiles, making compression and tiled GeoTIFF writing important; the local filesystem once reached 100% utilization, demonstrating that the pipeline must account for intermediate storage, not only final dataset size. Finally, Earth Engine authentication depends on external network services, so authentication is kept separate from per-lake processing and failures are recorded rather than silently producing incomplete data.

Because the project was developed by multiple team members, a database consolidation procedure was also designed: each developer's PostgreSQL dump is restored into a temporary database, validated, and merged into production, preserving PostGIS geometries and pgvector embeddings in a way CSV-based migration could not.

Chapter 6

Applied Analysis: Five Ramsar Wetlands and Reservoirs

6.1 Purpose and Site Selection

This chapter runs the system end to end on five Ramsar-designated sites, including inconclusive results: Chilika, Tso Morari, Maharana Pratap Sagar (Pong Dam, on the Beas), Tawa Reservoir, and Sasthamcotta. Ramsar designation does not by itself supply a checkable public condition report, so each site closes with a *Documented Condition* subsection that searches for one and reports the match honestly — ranging from Chilika’s renewed-weed-growth reports lining up with the measured signal, through Beas and Tawa (no published directional trend to check against), to Sasthamcotta, whose documented decline only *partially* agrees with the measured signal. Tso Morari additionally offers the chapter’s strongest independent corroboration via its Pangong Tso analog pairing (Section 7). The five sites, all with complete ten-year records, span a deliberately wide range of climate, altitude, and water-body type.

Table 6.1: The five Ramsar-designated sites analysed. All five have complete 2016–2025 embedding and feature coverage.

Site	State	Area	Elev.	Type
Chilika	Odisha	843.22	1	Brackish coastal lagoon
Tso Morari	Ladakh	141.25	4522	High-altitude cold desert
Tawa Reservoir	Madhya Pradesh	101.61	348	Irrigation reservoir (Ramsar site, 2024)
Maharana Pratap Sagar (Beas)	Himachal Pradesh	189.40	407	Himalayan-foothill reservoir (Pong Dam)
Sasthamcotta	Kerala	3.19	4	Freshwater lake, drinking-water reservoir

The spread — sea level to 4522 m, tropical estuary to Trans-Himalayan cold desert, two orders of magnitude in area — is deliberate: a system that only worked on lakes resembling one another would be exposed by it.

Every site was analysed under an identical protocol: a five-year query trajectory (2021–2025), a sliding-window search over all 828 other lakes, selection of the top five usable twins, delta extraction, fixed rank weights with per-year renormalization, and

classification at $\tau = 0.005$ (Chapter 3), with anchor year 2025 and forecasts covering 2026–2028.

6.2 Chilika Lagoon, Odisha

Chilika is Asia’s largest brackish-water lagoon and the largest lake in the cohort (843 km²), separated from the Bay of Bengal by a barrier spit.



Figure 6.1: Chilika’s lake boundary in the live application.

6.2.1 Retrieved Analogs

Table 6.2: Chilika — top five analogs (828 candidates, 146 ms).

#	Lake	State	Area	Raw cos.	Window
1	15704	Andhra Pradesh	12.28	0.99691	2017–2021
2	Govind Ballabh Pant	Uttar Pradesh	398.80	0.99646	2021–2025
3	177805	Gujarat	6.90	0.99554	2019–2023
4	15714	Telangana	27.56	0.99550	2018–2022
5	1509	Telangana	335.78	0.99536	2020–2024

Three of the five matched windows are *time-shifted*: the top analog matched on 2017–2021, so its 2022–2024 behaviour is already-observed evidence about a trajectory Chilika is only now traversing (Section 3.4).

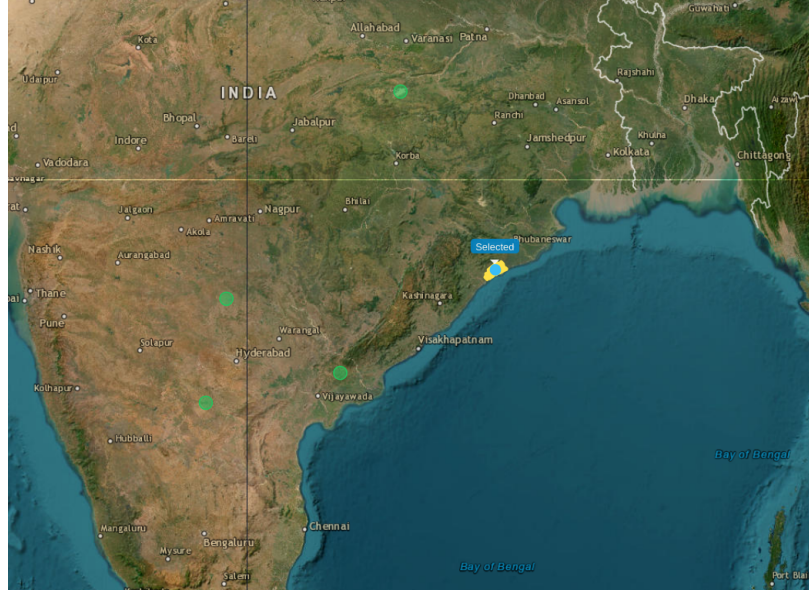


Figure 6.2: Chilika's retrieved analogs (Table 6.2) in the live application.

6.2.2 Measured History

Table 6.3: Chilika — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.0360	0.0323	0.0463	0.0518	0.0527	0.0415
NDVI	−0.0773	−0.1893	−0.1410	−0.1246	−0.1092	−0.1280
NDWI	0.1231	0.3170	0.2688	0.2258	0.2161	0.2441
Turbidity	−0.0463	−0.1359	−0.1328	−0.1041	−0.1095	−0.1198
Red-edge slope	−2.3037	−2.9020	−2.5017	−2.7318	−2.2062	−2.3865

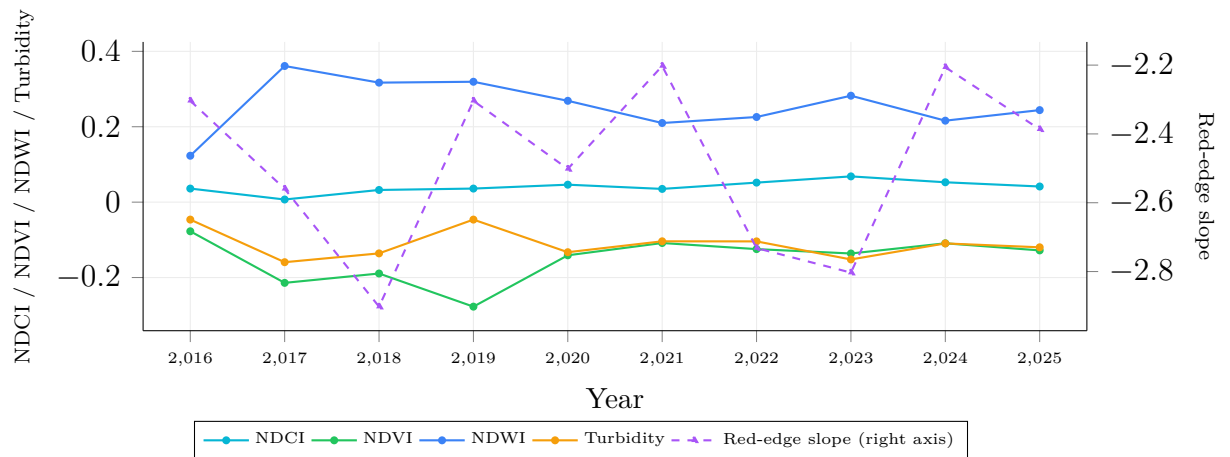


Figure 6.3: Chilika — measured index trajectory, 2016–2025.

The persistently negative NDVI and positive NDWI are the expected signature of a large, genuinely open water body.

6.2.3 Forecast

Table 6.4: Chilika — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.0415	→ (+0.0029, $n=5$)	→ (+0.0008, $n=4$)	↑ (+0.0391, $n=3$)
NDVI	−0.1280	↑ (+0.0253, $n=5$)	↑ (+0.0420, $n=4$)	↑ (+0.0730, $n=3$)
NDWI	0.2441	↓ (−0.0219, $n=5$)	↓ (−0.0578, $n=4$)	↓ (−0.0986, $n=3$)
Turbidity	−0.1198	→ (−0.0032, $n=5$)	↑ (+0.0219, $n=4$)	↑ (+0.0333, $n=3$)
Red-edge	−2.3865	↑ (+0.2800, $n=5$)	↑ (+0.3180, $n=4$)	↑ (+0.2312, $n=3$)

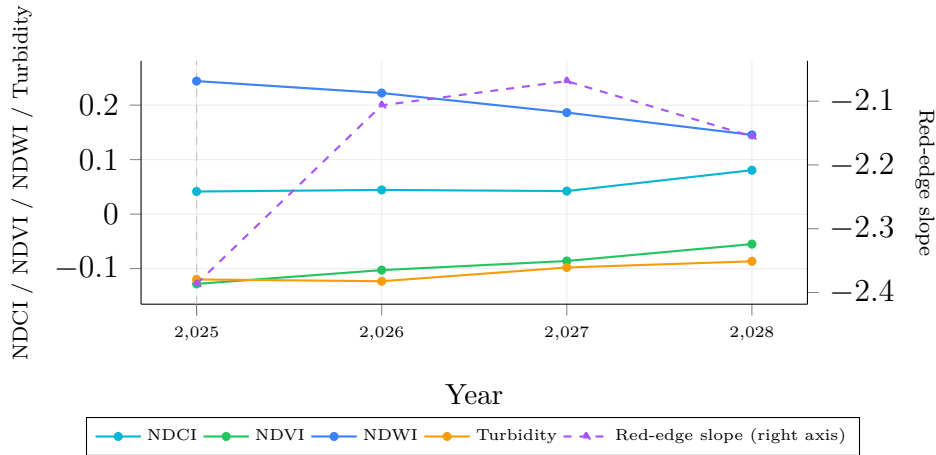


Figure 6.4: Chilika — projected index trajectory, 2025–2028.

6.2.4 Interpretation

The forecast is internally coherent: NDVI rises while NDWI falls by a comparable magnitude every year (both share B8), and rising red-edge slope and turbidity reinforce the same reading — **gradual vegetative encroachment and reduction of open water** — with the declining twin count (5, 4, 3) signalling 2028 should be read more cautiously than 2026.

6.2.5 Documented Condition

Chilika’s weed cover exploded from ~ 20 to ~ 685 km² between 1972 and 2000 as siltation reduced salinity, prompting the 2000 sea-mouth reopening that sharply cut

weed cover.¹ Recent reporting describes renewed weed growth interfering with fishing — consistent with the gradual encroachment signal measured here, a real but far milder trend than the pre-2000 crisis.

6.3 Maharana Pratap Sagar (Beas), Himachal Pradesh

Maharana Pratap Sagar — the reservoir impounded by the Pong Dam on the Beas River, a Ramsar site since 2002 and a major wintering ground for migratory waterfowl in the western Himalayan foothills — is identified in the dataset by its source river, **Beas** (lake_id 1423). Unlike the natural wetlands elsewhere in this chapter, it is a dam-regulated reservoir, so its surface extent and shoreline vegetation respond to reservoir operations (drawdown and refill) as much as to natural hydrology or ecological succession.

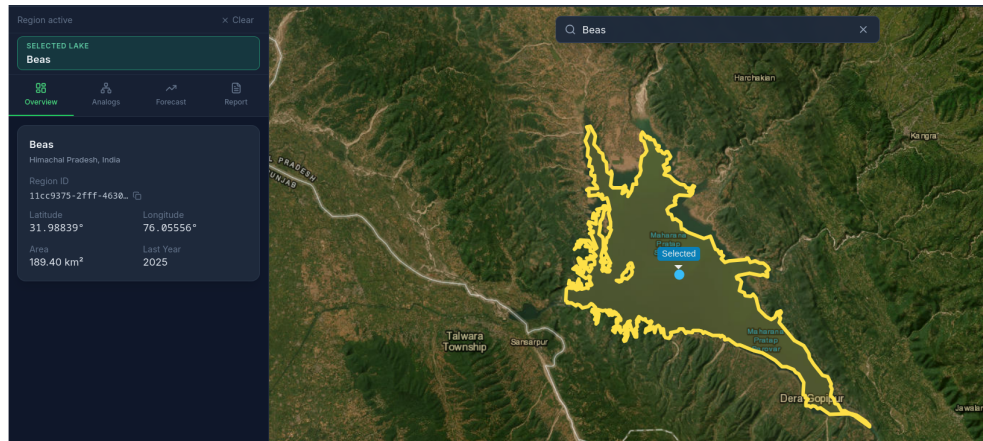


Figure 6.5: Beas (Maharana Pratap Sagar)’s lake boundary in the live application.

6.3.1 Retrieved Analogs

Table 6.5: Beas — top five analogs (828 candidates).

#	Lake	State	Area	Raw cos.	Window
1	Gandhisagar Reservoir	Madhya Pradesh	523.51	0.99884	2021–2025
2	15581	Chhattīsgarh	28.46	0.99824	2021–2025
3	15672	Telangana	49.45	0.99816	2020–2024
4	178210	Chhattīsgarh	8.75	0.99812	2020–2024
5	15727	Andhra Pradesh	19.52	0.99810	2016–2020

¹Chilika Development Authority site history; Down To Earth, “Chilika: a lake in limbo”, <https://www.downtoearth.org.in/environment/chilika-a-lake-in-limbo-30053>.

The top analog is itself another large dam reservoir (Gandhisagar, on the Chambal), sharing Beas’s man-made, regulated character. The two top-ranked analogs’ matched windows both end at 2025 (the anchor year), so neither has an observed future to contribute; the twins actually used for forecasting are drawn from further down the ranked pool.

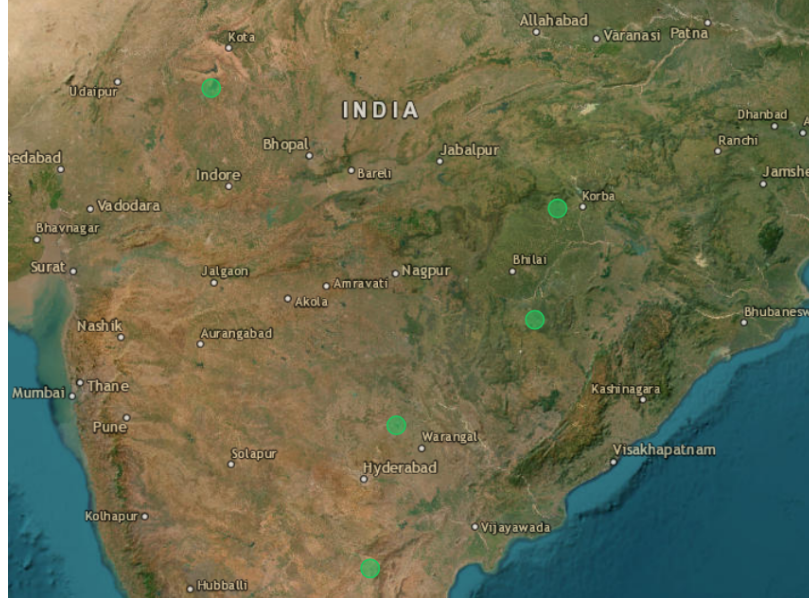


Figure 6.6: Beas’s retrieved analogs (Table 6.5) in the live application.

6.3.2 Measured History

Table 6.6: Beas — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	−0.0039	−0.0061	−0.0231	0.0589	−0.0021	0.0208
NDVI	−0.0996	−0.0818	−0.1595	0.0749	−0.1185	−0.2293
NDWI	0.2673	0.2647	0.3680	0.1325	0.2746	0.3733
Turbidity	−0.1723	−0.1870	−0.2215	−0.2054	−0.1614	−0.1574
Red-edge slope	−0.8025	−0.7993	−0.5696	0.2272	−0.9782	−2.8268

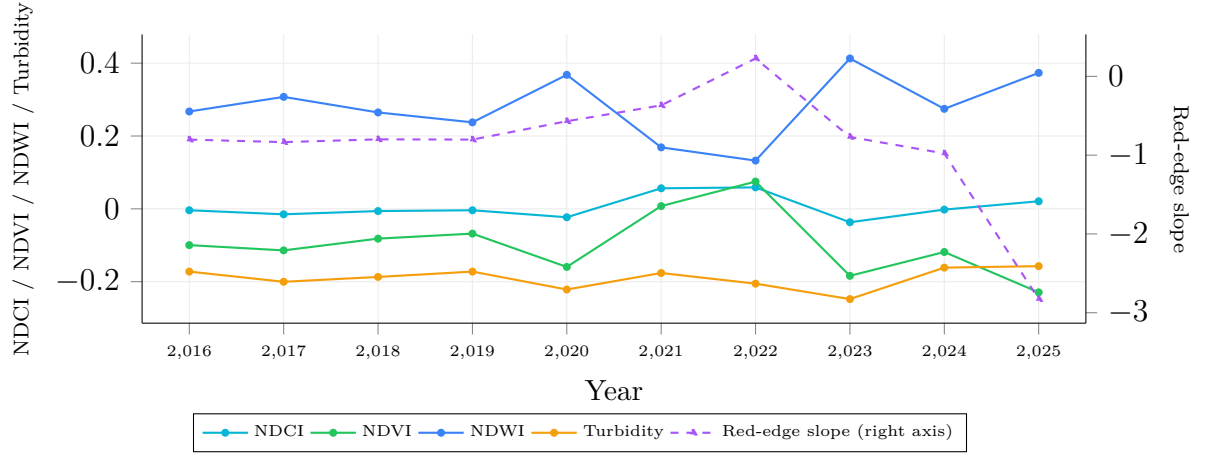


Figure 6.7: Beas — measured index trajectory, 2016–2025.

Beas shows no sustained directional trend of the kind seen at Wular in the earlier version of this chapter: NDVI oscillates between roughly -0.23 and $+0.07$ and NDWI between 0.13 and 0.41 with no monotone drift, consistent with a reservoir whose surface extent is driven by dam operations (seasonal drawdown and refill for irrigation and hydropower) rather than by a one-way ecological process. The red-edge slope is the most volatile series measured anywhere in this study, swinging from $+0.23$ in 2022 to -2.83 in 2025 — a signal to read cautiously rather than as a pigment trend, since it is exactly the kind of index a fluctuating reservoir shoreline can perturb independently of vegetation health.

6.3.3 Forecast

Table 6.7: Beas — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.0208	↓ (-0.0276 , $n=5$)	? ($n=1$)	? ($n=1$)
NDVI	-0.2293	↑ ($+0.0123$, $n=5$)	? ($n=1$)	? ($n=1$)
NDWI	0.3733	→ (-0.0006 , $n=5$)	? ($n=1$)	? ($n=1$)
Turbidity	-0.1574	↓ (-0.0118 , $n=5$)	? ($n=1$)	? ($n=1$)
Red-edge	-2.8268	↑ ($+0.5993$, $n=5$)	? ($n=1$)	? ($n=1$)

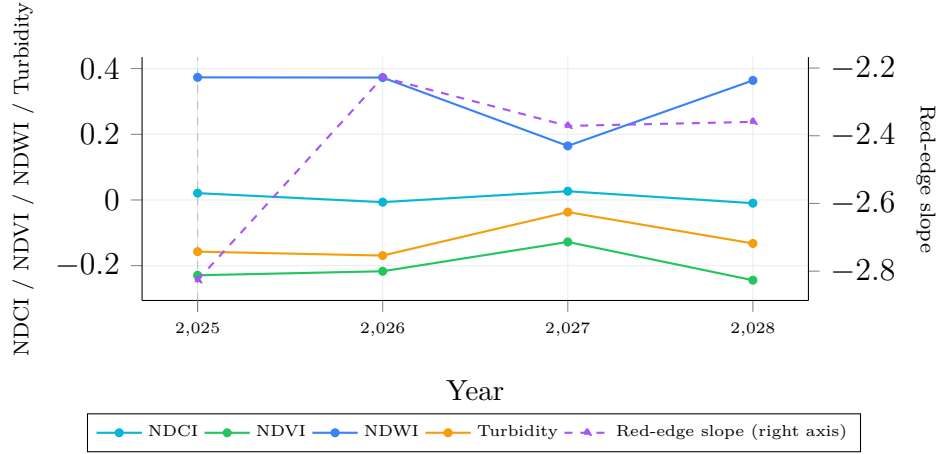


Figure 6.8: Beas — projected index trajectory, 2025–2028.

6.3.4 Interpretation

Beas shows the most confidence degradation in the study: four of its five twins matched on windows ending in 2024, leaving only 2026 ($h = 1$) with all five contributing ($n = 5, 1, 1$), so 2027/2028 are correctly reported as *uncertain* rather than resting on one lake’s trajectory. The 2026 read is mild: NDCI down, NDVI and red-edge up, NDWI essentially flat.

6.3.5 Documented Condition

Public documentation of Maharana Pratap Sagar (Pong Dam Lake) centres on its status as a Ramsar-designated bird sanctuary and a multi-use reservoir combining fishery and hydroelectric generation.² No published account of a directional trend specific to this reservoir was found — consistent with, but not confirmation of, the oscillating, no-sustained-trend signal measured here.

6.4 Tawa Reservoir, Madhya Pradesh

Tawa Reservoir is a large irrigation and flood-control reservoir on the Tawa River, a tributary of the Narmada, in Madhya Pradesh’s Narmadapuram district, within the Satpura Tiger Reserve — notified as a Ramsar site of international importance in 2024. It is included here as a second dam-regulated freshwater comparison point alongside Beas (Section 6.3.3), replacing Loktak, which turned out not to be identifiable with confidence in the active processing cohort (there is no lake named “Loktak” in the underlying HydroLAKES-derived table; the earlier version of this chapter’s identification of lake_id 15510 with Loktak was an unverified inference from area and state alone).

²Ramsar site designation, 2002 (criteria 5 and 8); overview at https://en.wikipedia.org/wiki/Maharana_Pratap_Sagar.

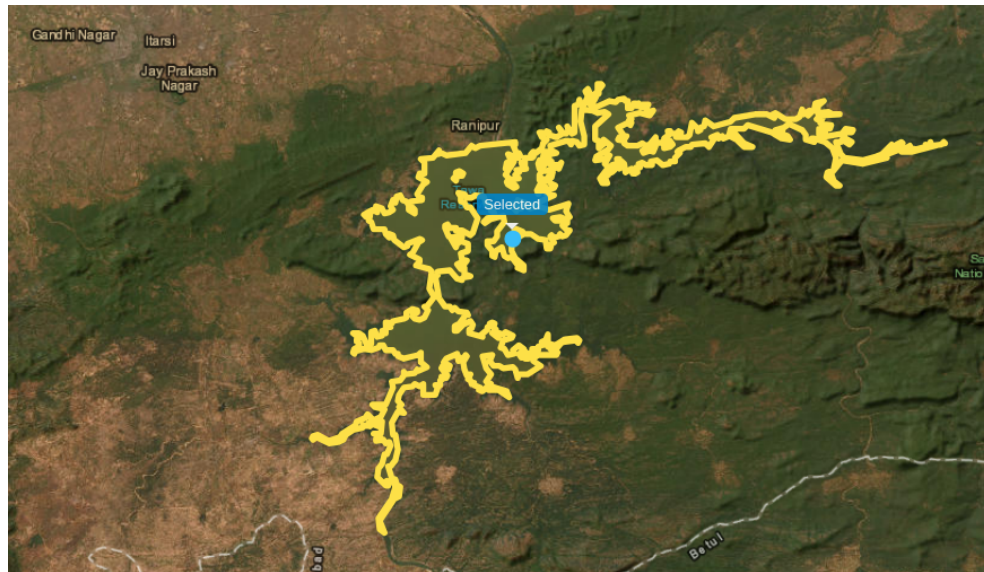


Figure 6.9: Tawa Reservoir’s lake boundary in the live application.

6.4.1 Retrieved Analogs

Table 6.8: Tawa Reservoir — top five analogs (828 candidates).

#	Lake	State	Area	Raw cos.	Window
1	15544	Madhya Pradesh	15.03	0.99936	2021–2025
2	15634	Maharashtra	15.54	0.99931	2019–2023
3	15616	Maharashtra	39.91	0.99926	2017–2021
4	178333	Telangana	5.12	0.99921	2019–2023
5	15577	Madhya Pradesh	36.00	0.99913	2021–2025

The top and fifth-ranked analogs both match on windows ending at 2025 and so carry no observed future, so the forecast twins used below are drawn from ranks 2–4 plus two further candidates.



Figure 6.10: Tawa Reservoir’s retrieved analogs (Table 6.8) in the live application.

6.4.2 Measured History

Table 6.9: Tawa Reservoir — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	−0.0874	0.0479	0.0011	0.0009	−0.0107	−0.0423
NDVI	−0.2939	0.0681	−0.1323	−0.0890	−0.1606	−0.1918
NDWI	0.4824	0.0059	0.2650	0.2450	0.3178	0.3701
Turbidity	−0.2197	−0.0739	−0.1375	−0.1595	−0.1657	−0.1920
Red-edge slope	−1.3090	0.2360	−1.1605	−0.5719	−0.9505	−0.9585

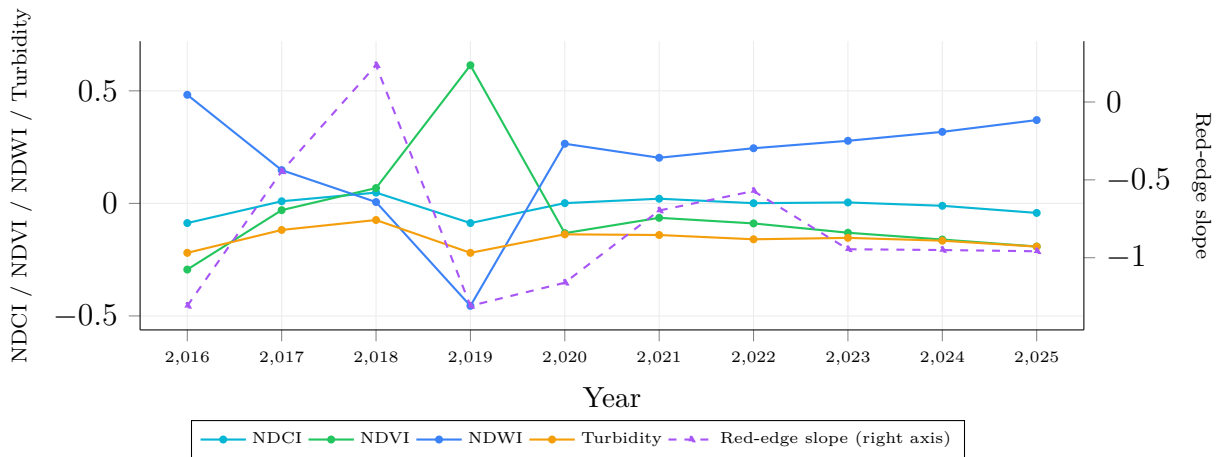


Figure 6.11: Tawa Reservoir — measured index trajectory, 2016–2025.

Like Beas, Tawa shows no sustained one-way trend — every index oscillates across the decade rather than drifting monotonically, consistent with a managed reservoir

where surface water extent tracks irrigation releases and monsoon refill rather than a directional ecological process. 2018 stands out as a local extremum across all five indices simultaneously (least negative NDCI/NDVI, lowest NDWI, least negative turbidity, only positive red-edge slope value in the series) — most plausibly a single unusually dry or low-water year rather than a data artifact, though it was not independently cross-checked against reservoir operation records.

6.4.3 Forecast

Table 6.10: Tawa Reservoir — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	−0.0423	↑ (+0.0129, $n=5$)	↓ (−0.0158, $n=5$)	↑ (+0.0200, $n=2$)
NDVI	−0.1918	↑ (+0.0207, $n=5$)	↓ (−0.0071, $n=5$)	↑ (+0.0056, $n=2$)
NDWI	0.3701	↓ (−0.0420, $n=5$)	↑ (+0.0132, $n=5$)	↓ (−0.0348, $n=2$)
Turbidity	−0.1920	↑ (+0.0249, $n=5$)	↓ (−0.0075, $n=5$)	↑ (+0.0316, $n=2$)
Red-edge	−0.9585	↓ (−0.0282, $n=5$)	↑ (+0.0673, $n=5$)	↓ (−0.1696, $n=2$)

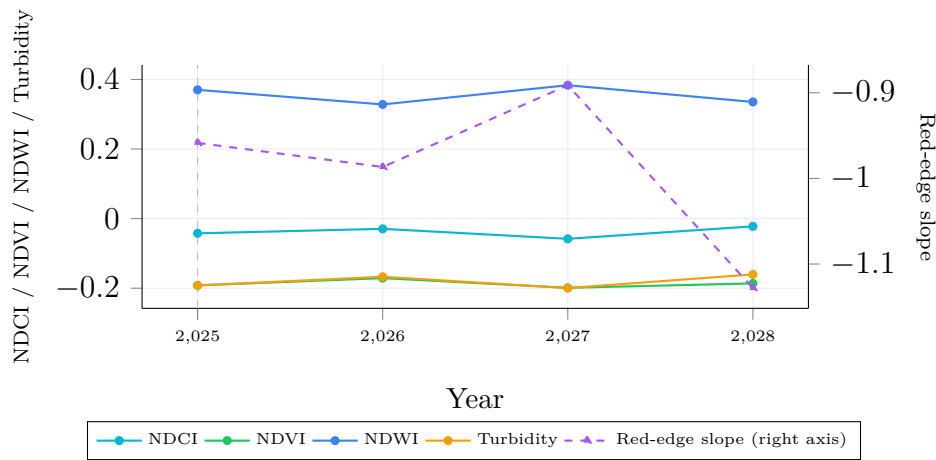


Figure 6.12: Tawa Reservoir — projected index trajectory, 2025–2028.

6.4.4 Interpretation

Every index alternates direction year over year, an oscillatory forecast that mirrors the oscillation already visible in the measured decade rather than projecting a new one-way trajectory. Confidence holds at $n = 5$ for 2026–2027 before dropping to $n = 2$ in 2028; the 2028 red-edge score (−0.1696) is the largest-magnitude cell in this table but rests on only two twins, so it should be read as suggestive, not calibrated.

6.4.5 Documented Condition

Tawa Reservoir was notified as a Ramsar site in August 2024 on the strength of its biodiversity: 57 fish species, 21 mammals, and over 280 migratory bird species.³ As with Beas, no documented one-way ecological trend was found to check the measured oscillation against — the designation reflects current ecological richness rather than a published condition trajectory.

6.5 Tso Morari, Ladakh

Tso Morari is a high-altitude brackish lake at 4522 m in the Trans-Himalayan cold desert — the cohort’s environmental extreme, with minimal vegetation and glacial meltwater input.



Figure 6.13: Tso Morari’s lake boundary in the live application.

³MoEFCC Ramsar notification, August 2024; Ramsar Sites Information Service, <https://rsis.ramsar.org/ris/2547>.

6.5.1 Retrieved Analogs — The Strongest Validation

Table 6.11: Tso Morari — top five analogs (828 candidates, 160 ms).

#	Lake	State	Area	Raw cos.	Window
1	Pangong Tso	Ladakh	285.46	0.99720	2021–2025
2	178351	Maharashtra	8.52	0.98816	2021–2025
3	15661	Maharashtra	10.12	0.98781	2021–2025
4	178391	Maharashtra	3.06	0.98769	2021–2025
5	15828	Kerala	26.07	0.98599	2021–2025

This is the single most convincing retrieval result in the study. Out of 828 candidates spanning 26 states, the top-ranked analog for Tso Morari is **Pangong Tso** — the other great high-altitude brackish lake of Ladakh, roughly 120 km away — recovered from reflectance alone, with no coordinates or lake-type metadata supplied. The score gap makes it stronger still: 0.99720 against rank 2’s 0.98816, five times the spread between ranks 2 and 5, indicating the embedding space genuinely isolates high-altitude cold-desert lakes as a distinct class.

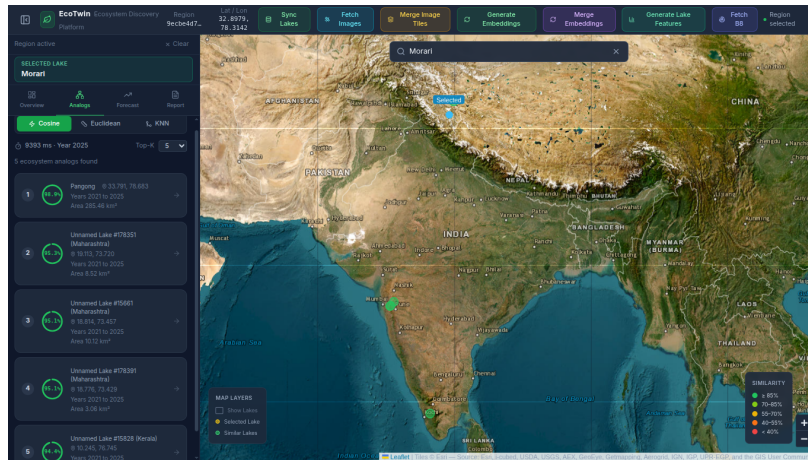


Figure 6.14: Tso Morari’s ranked analogs in the live application; Pangong returns at rank 1, confirming Table 6.11.

6.5.2 Measured History

Table 6.12: Tso Morari — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	−0.0396	−0.0212	−0.0417	−0.0450	−0.0337	−0.0376
NDVI	−0.1584	−0.1030	−0.1000	−0.0926	−0.0501	−0.0510
NDWI	0.5315	0.2805	0.4086	0.3934	0.2849	0.3371
Turbidity	−0.4074	−0.1828	−0.3217	−0.3121	−0.2382	−0.2912
Red-edge slope	−0.0893	−0.2676	−0.1009	−0.0617	−0.0007	−0.0446

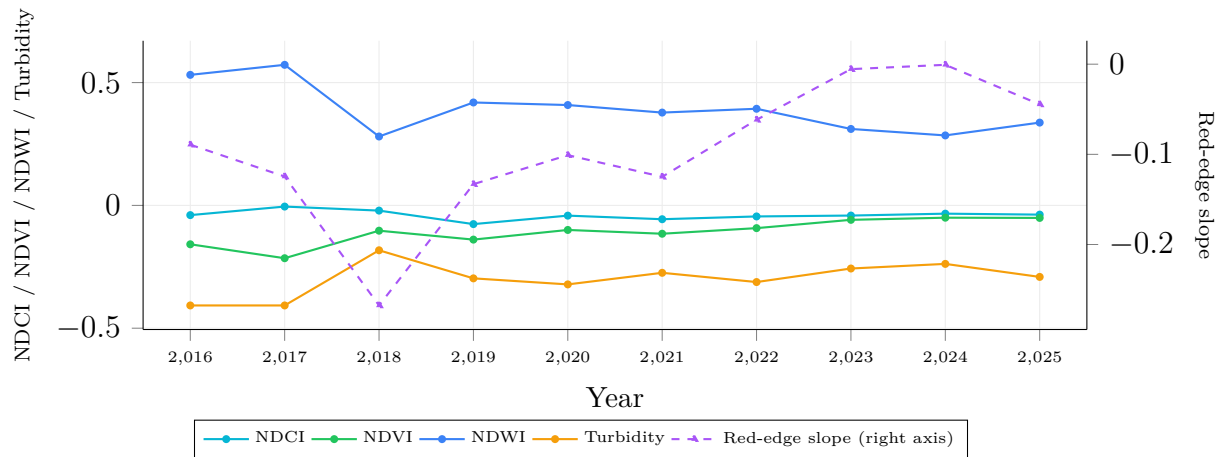


Figure 6.15: Tso Morari — measured index trajectory, 2016–2025.

Tso Morari records the **highest NDWI** (0.53) and **most negative turbidity** (−0.41) in the study — the signature of exceptionally clear, vegetation-free, glacier-fed water, and the opposite extreme from Sasthamcotta’s vegetation-heavy 2016 reading (Section 6).

6.5.3 Forecast

Table 6.13: Tso Morari — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	−0.0376	↑ (+0.0204, $n=5$)	→ (+0.0046, $n=5$)	↓ (−0.0057, $n=5$)
NDVI	−0.0510	↑ (+0.0310, $n=5$)	↑ (+0.0262, $n=5$)	↓ (−0.0145, $n=5$)
NDWI	0.3371	↓ (−0.0482, $n=5$)	↓ (−0.0260, $n=5$)	↓ (−0.0062, $n=5$)
Turbidity	−0.2912	↑ (+0.0204, $n=5$)	↓ (−0.0058, $n=5$)	↑ (+0.0172, $n=5$)
Red-edge	−0.0446	↑ (+0.0499, $n=5$)	↑ (+0.2908, $n=5$)	↓ (−0.0522, $n=5$)

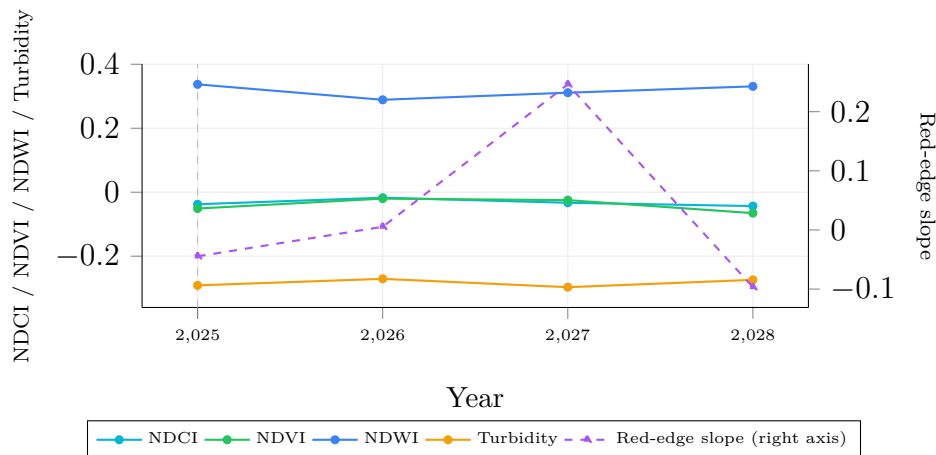


Figure 6.16: Tso Morari — projected index trajectory, 2025–2028.

At the time this screenshot was captured, the live application’s Forecast panel (Figure 6.17) agreed with the then-current (B7-proxy) forecast on **direction for all fifteen cells**, despite drawing on 15 contributing twins rather than the 5 used offline — agreement despite a threefold difference in twin-pool size was evidence the classification was not an artifact of which handful of twins happened to be selected. That agreement has not yet been re-verified against the B8-based numbers in Table 6.13, since 2028’s NDWI cell changed direction ($\rightarrow$ became $\downarrow$) between the two implementations; a fresh capture is listed as follow-up work.

6.5.4 Interpretation

Tso Morari is the only site where all five twins contribute to all three forecast years ($n = 5$ throughout), the best-supported forecast in the study. The projection is mild — modest NDVI/NDCI increases, NDWI declining across all three years without reversal — appropriately undramatic for an ecosystem with no significant vegetation dynamic.

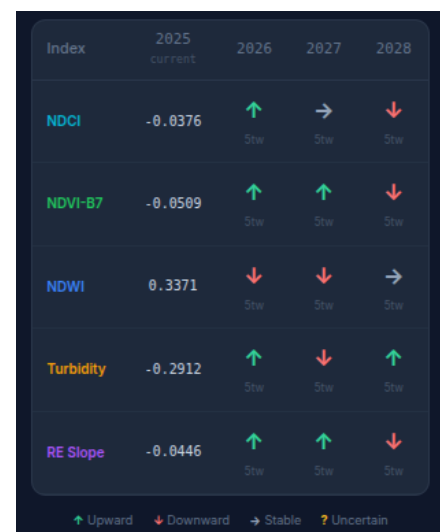


Figure 6.17: Tso Morari’s forecast summary under the earlier B7-proxy implementation, predating the switch to true-NIR B8 (Table 6.13).

6.5.5 Documented Condition

The literature on Tso Moriri attributes its water-level fluctuations primarily to within-year forcing (snowmelt and glacial meltwater against winter precipitation and evaporation) rather than a short secular trend, while flagging accelerating glacier mass loss across the wider Trans-Himalayan region as a longer-horizon concern.⁴ The near-

⁴Extreme seasonal water-level changes in high-altitude Himalayan lakes, *Scientific Reports* (2023), <https://www.nature.com/articles/s41598-023-37667-z>.

decade stability measured here is consistent with a system dominated by within-year forcing, but cannot speak to the slower glacial retreat documented at the regional scale.

6.6 Sasthamcotta Lake, Kerala

Sasthamcotta is Kerala’s largest freshwater lake and the principal drinking-water source for Kollam district, a Ramsar site since 2002, freshwater despite lying only a few kilometres from the coast and close to the tidally-influenced Vembanad backwater system it replaces in this chapter. It is identified in the dataset as Unnamed Lake #179921 (Kerala), lake_id 179921.

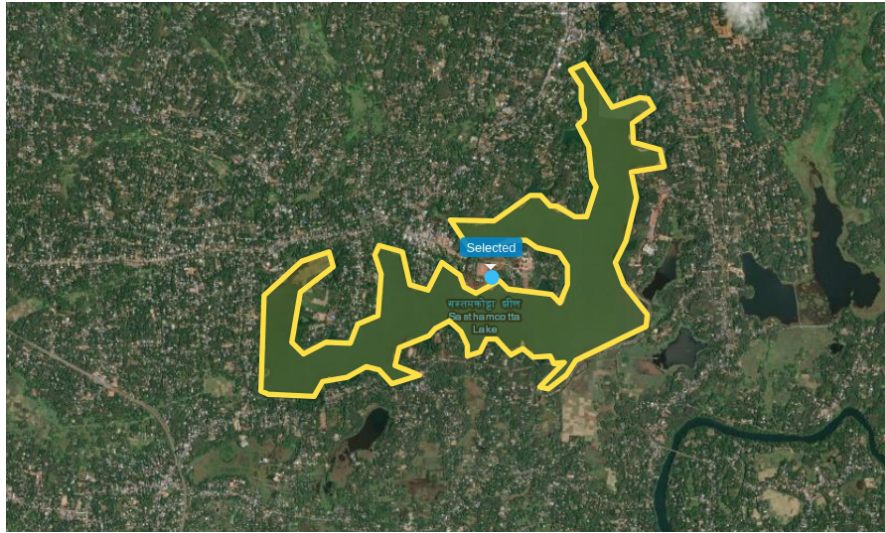


Figure 6.18: Sasthamcotta’s lake boundary in the live application.

6.6.1 Retrieved Analogs

Table 6.14: Sasthamcotta — top five analogs (828 candidates).

#	Lake	State	Area	Raw cos.	Window
1	179945	Kerala	2.50	0.99784	2019–2023
2	179981	Kerala	2.59	0.99777	2016–2020
3	180015	Tamil Nadu	2.08	0.99754	2017–2021
4	179746	Kerala	4.16	0.99709	2019–2023
5	15611	Maharashtra	11.75	0.99698	2019–2023

Unlike every other site in this chapter, all five top-ranked analogs here are directly usable as forecast twins — none matches on a window ending at the 2025 anchor year, so no substitution from further down the ranked pool was needed. Three of the five

are other small Kerala water bodies, the closest geographic clustering of analogs seen in this study.

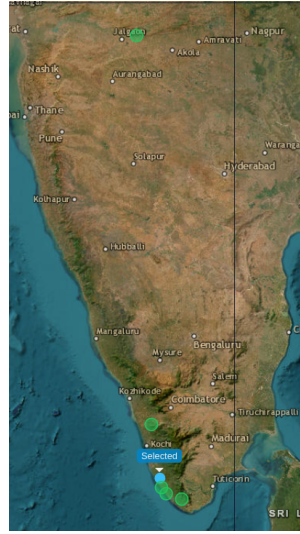


Figure 6.19: Sasthamcotta’s retrieved analogs (Table 6.14) in the live application.

6.6.2 Measured History

Table 6.15: Sasthamcotta — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.1957	0.0952	0.0823	0.0793	0.0928	0.0865
NDVI	0.4520	0.2606	0.1992	0.1006	0.1558	0.1816
NDWI	−0.2769	0.0128	0.0172	0.1487	0.1324	0.0731
Turbidity	−0.2002	−0.2725	−0.2157	−0.2456	−0.2824	−0.2513
Red-edge slope	4.9975	2.6302	1.8539	0.7420	1.0838	1.5259

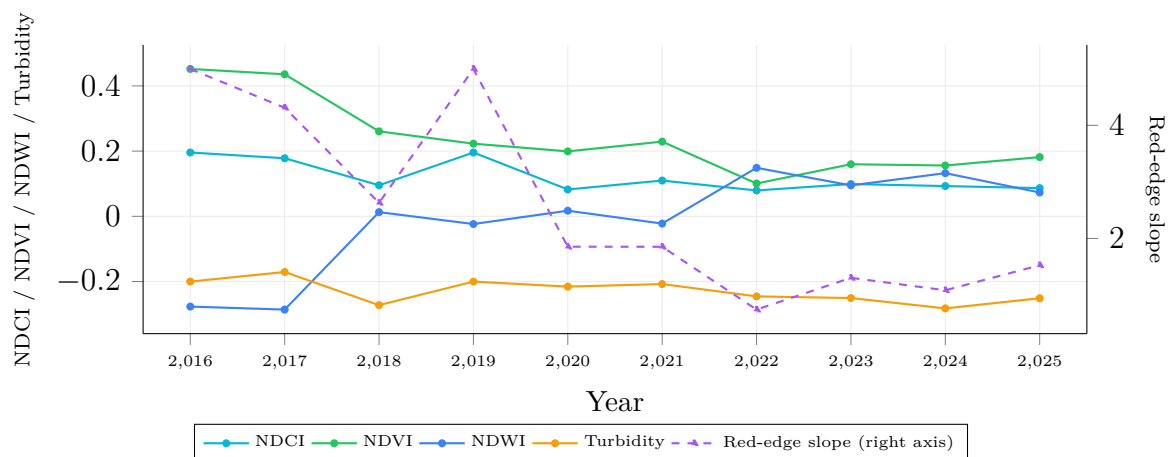


Figure 6.20: Sasthamcotta — measured index trajectory, 2016–2025.

Sasthamcotta’s measured decade shows a clear, largely monotone trend: NDVI falls by more than half ($0.452 \rightarrow 0.182$), the red-edge slope falls roughly 3.3-fold ($5.00 \rightarrow 1.53$), and NDWI moves from strongly negative to positive ($-0.277 \rightarrow 0.073$) — a decline in vegetation vigor and pigment signal alongside a widening open-water signature, most pronounced between 2016 and 2020 and comparatively stable since.

6.6.3 Forecast

Table 6.16: Sasthamcotta — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.0865	$\rightarrow (+0.0009, n=5)$	$\uparrow (+0.0088, n=5)$	$\uparrow (+0.0131, n=2)$
NDVI	0.1816	$\downarrow (-0.0083, n=5)$	$\rightarrow (-0.0032, n=5)$	$\downarrow (-0.0502, n=2)$
NDWI	0.0731	$\rightarrow (-0.0046, n=5)$	$\rightarrow (+0.0028, n=5)$	$\uparrow (+0.0519, n=2)$
Turbidity	-0.2513	$\uparrow (+0.0096, n=5)$	$\rightarrow (+0.0027, n=5)$	$\downarrow (-0.0059, n=2)$
Red-edge	1.5259	$\downarrow (-0.0658, n=5)$	$\downarrow (-0.1167, n=5)$	$\downarrow (-0.5053, n=2)$

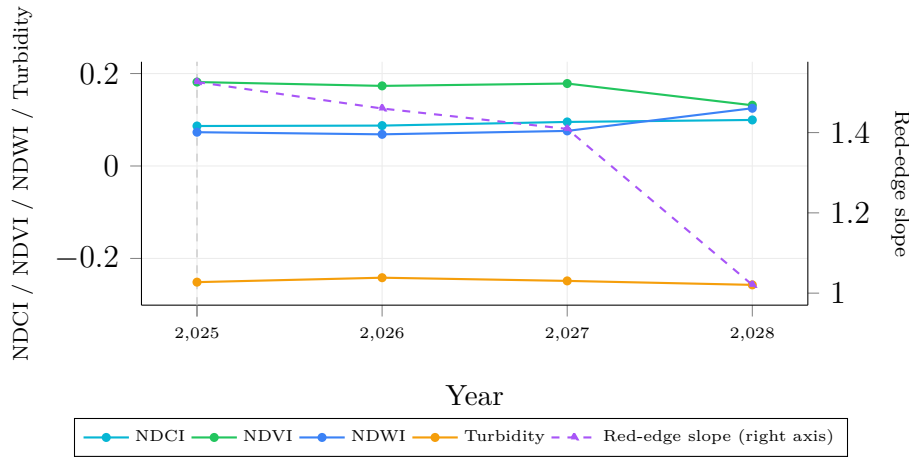


Figure 6.21: Sasthamcotta — projected index trajectory, 2025–2028.

6.6.4 Interpretation

The forecast projects a continuation of the measured decade-long trend: NDVI and red-edge slope keep declining in all three years while NDWI holds flat or rises, directionally consistent with the 2016–2025 history. Confidence follows the familiar pattern of this chapter, $n = 5, 5, 2$, with 2028’s red-edge score (-0.5053) resting on only two twins and thus the least reliable number here; 2026 and 2027 are the rows to trust.

6.6.5 Documented Condition

Sasthamcotta’s decline is extensively documented: elevated COD/BOD and turbidity, shoreline encroachment removing an estimated 15% of the lake’s perimeter, illegal

sand extraction lowering water levels, and over 40% of native fish species disappeared in two decades.⁵ This is a genuinely degrading system by every published account, but the measured signal only *partially* agrees: NDVI and red-edge decline as expected, but NDWI *rises* over the decade, the opposite of what a shrinking lake would suggest. A plausible reconciliation is that encroachment and sand mining chiefly erode the lake’s perimeter and depth — under-represented by a coverage-weighted mean over the remaining analysis cells — while the NDVI/NDCI decline more likely reflects falling phytoplankton than the water-hyacinth pressure seen on other Kerala wetlands (which would push NDVI up, not down). The measured signal and public record should be read together here, each qualifying the other, rather than one confirming the other.

6.7 Cross-Site Synthesis

Table 6.17: Cross-site summary.

Site	Best raw	Dominant measured signal	Conf.
Chilika	0.99691	Mild vegetative encroachment	5,4,3
Maharana Pratap Sagar (Beas)	0.99884	No sustained trend; reservoir-driven oscillation	5,1,1
Tawa Reservoir	0.99936	Oscillatory, no net trend	5,5,2
Tso Morari	0.99720	Stable clear cold-desert water	5,5,5
Sasthamcotta	0.99784	Decade-long NDVI/pigment decline	5,5,2

Four findings emerge from analysing the sites together. The embedding space encodes real ecological structure: the Tso Morari–Pangong pairing, recovered from reflectance alone with a score gap five times the typical inter-rank spread, is not explicable as coincidence. Condition dominates geography in analog retrieval: Beas’s top analogs are reservoirs hundreds of kilometres away in entirely different climate zones, matched on hydrological state rather than location, meaning the map view alone can mislead without the index history. The index set discriminates across the full range, from Tso Morari’s near-zero vegetation to Sasthamcotta’s NDVI of 0.45, preserved even after the B7-to-B8 switch (Section 4.8). Forecast confidence is bounded by data recency, not model quality: every degraded forecast traces to twins whose matched windows end near 2025, a limitation that self-corrects as further years are ingested. Finally, the system declines to answer when it should: across 75 forecast cells (5 sites $\times$ 5 indices $\times$ 3 years), the outcomes divide as shown in Table 6.18.

⁵Water Quality Assessment of Sasthamcotta Lake, Kollam, Kerala, <https://www.researchgate.net/publication/323628043>.

Table 6.18: Distribution of forecast cells across sites.

Classification	Cells	Share
Up ($\uparrow$)	30	40.0%
Down ($\downarrow$)	25	33.3%
Stable ($\rightarrow$)	10	13.3%
Uncertain (?)	10	13.3%
Total	75	100%

Thirteen percent of cells returned *uncertain* rather than a direction: a method that always answers cannot be distinguished from one that guesses, so these ten abstentions are evidence the confidence machinery is load-bearing. The near-balance of up and down classifications is also reassuring against directional bias.

Chapter 7

Discussion

7.1 Strengths and Limitations

The system’s central strengths are scalable offline preprocessing executed to completion with zero residual failures (Chapter 5); trajectory-based retrieval that matches ecological *condition* rather than single-year appearance, demonstrated by the Tso Morari–Pangong pairing (Chapter 6); and a decoder-free, fully auditable forecasting layer that reports *uncertain* rather than guessing when evidence is thin (13.3% abstention across 75 forecast cells).

Its limitations are structural. The ten-year window is not uniformly complete, and cloud cover reduces valid Sentinel-2 observations. A high embedding similarity does not guarantee ecological equivalence — only the ranking is meaningful, not the compressed absolute score — and every result is conditional on the fixed configuration of Table 5.1. A twin whose matched window ends near 2025 has little observed future to transfer, which thinned Beas’s and Tawa/Sasthamcotta’s later-horizon forecasts to one or two twins (Chapter 6) — a consequence of dataset extent, not a flaw in the classification rule, which correctly reported *uncertain* where warranted. Finally, the documented-condition checks are corroborating evidence, not calibrated validation, and their strength varies by site: Chilika’s signal lines up with independently reported weed growth, Sasthamcotta’s decline only partially agrees with the measured signal, Beas and Tawa have no published trend to check against, and Tso Morari’s strongest corroboration is internal to the retrieval method; formal hindcasting (Section 3.5.2) is the next step precisely because it does not require external data.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

EcoTwin evolved from a conceptual ecosystem-comparison idea into a complete, fully executed geospatial retrieval and forecasting system: HydroLAKES and Sentinel-2 feed a buffered, tiled acquisition pipeline into Prithvi embeddings, coverage-pooled to lake level and indexed in pgvector, from which a five-year sliding-window trajectory search ranks analogs and a decoder-free forecasting layer transfers their measured index changes onto the target with a full audit trail — executed to completion over 829 lakes, ten years, and zero outstanding failures.

Chapter 6 demonstrated the pipeline on five Ramsar wetlands, returning a striking unsupervised validation (Tso Morari’s top analog is Pangong Tso, the only other high-altitude Ladakhi lake in the cohort), a measured encroachment signal at Chilika consistent with independently reported weed growth, a decade-long decline at Sasthamcotta that only partially agrees with its own documented degradation, and a forecasting method that correctly declined to predict in 10 of 75 cells rather than manufacturing a confident answer. The project’s most significant contribution is the integration of geospatial data engineering, foundation-model inference, vector retrieval, temporal analysis, and visualization into one reproducible workflow carried through to applied results.

8.2 Future Work

Priority items are executing the hindcasting protocol once further years extend the dataset past its current single-cutoff limitation; extending ingestion beyond 2025 to resolve *uncertain* cells; adding independent field-ecological validation; and migrating trajectory search to pgvector’s native ANN operators, since the current brute-force scan will not remain interactive as the cohort grows. Further directions include calibrating the similarity display scale against labelled ecosystem pairs, learned-weighted pooling in place of coverage-weighted mean, growing the cohort beyond 829 lakes, and deploying shared production infrastructure.

8.3 Final Remarks

EcoTwin’s central idea is to use historically observed ecosystems as computationally retrievable analogs rather than treating every ecosystem as an isolated forecasting

problem. The Ramsar case studies show that this idea, carried through a fully processed cohort, produces results that are specific, checkable against independent knowledge where it exists, and honest about the limits of what it can currently claim where the evidence ran out.

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